

INSTITUTO SUPERIOR TÉCNICO

DEPARTAMENTO DE ENGENHARIA INFORMÁTICA

FORENSICS CYBER-SECURITY

MEIC, METI

Lab Assignment III

OPERATION DEEPFISH – Stage III

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Introduction

This assignment concludes the investigation into the "OPERATION DEEPFISH" case. In the previous stage, you analyzed the disk and memory images from two computers belonging to João Musk, the main suspect behind the phishing campaign discovered in Lab 1. The investigation now shifts its focus to the computer network of AMA (Agência para a Modernização Administrativa), the public agency responsible for the AMÉLIA platform. Evidence discovered on João Musk's computers provides a further investigative lead concerning the disclosure of information from AMA and possible manipulation of AMÉLIA and irregularities in public procurement. This assignment aims to investigate this new lead by examining forensic material collected from AMA's infrastructure. Your objectives are to establish the authenticity of the recovered material, reconstruct its circulation, assess the allegations it contains, and determine what the evidence supports concerning the individuals involved. As in previous assignments, we recommend conducting all analyses within a Kali Linux virtual machine.

Scenario presentation

The forensic analysis of João Musk's Windows laptop and Linux workstation provided decisive evidence confirming his authorship of the phishing campaign against the IST community. His computers contained a backup of the fake Fenix website, records of its development and deployment, and evidence of access to the credentials collected by the phishing server. Browser and command histories, together with Discord conversations, indicated that João had used stolen teaching assistant credentials to obtain Língua Natural exam materials. During a second interview, João admitted responsibility for the campaign and confirmed that his objective had been to obtain those materials through Tiago Sintaxe's academic account.

The investigation also uncovered material unrelated to the theft of exam materials. João's Linux workstation contained copies of the files carrying the hidden material recovered from the phishing website in the previous assignments. The hidden files had been imported from a USB drive and concealed within personal files. Subsequent repository synchronization and merging incorporated these files into the phishing website's repository, from which they were inadvertently published during deployment.

Recovered Discord conversations showed exchanges between João, known online as Johnny, and a contact using the nickname `DuckyDonald`. This contact had arranged for João to collect a USB drive from a locker at Alameda Metro station and asked him to keep its contents safe. In a later conversation, `DuckyDonald` identified himself as Tiago Sintaxe, an AMA Data Science intern working on the evaluation of AMÉLIA's retrieval-augmented generation (RAG) components and a teaching assistant for the Língua Natural course at IST, whose academic credentials João had stolen.

The examination of João's computers, the recovered material, and the Discord communications provided a further lead concerning possible irregularities involving AMÉLIA and public procurement. In light of these findings, the Polícia Judiciária opened a new case focused on investigating potential ongoing fraud at AMA. A judicial warrant was issued authorizing a forensic inspection of AMA's IT infrastructure, with particular focus on the systems most likely to contain traces of data leakage.

You have once again been asked to assist the Polícia Judiciária with this investigation. Specifically, your assignment is to examine the preserved network traces from AMA and find evidence involving this case. You have been briefed on AMA's network and provided with relevant forensic material. Figure 1 presents a reconstruction of the infrastructure at the time of evidence collection, showing the office subnet (`10.10.14.0/24`), the datacenter subnet (`10.10.20.0/24`), and their connection to external networks through a gateway with the external IP address `185.77.23.1`.

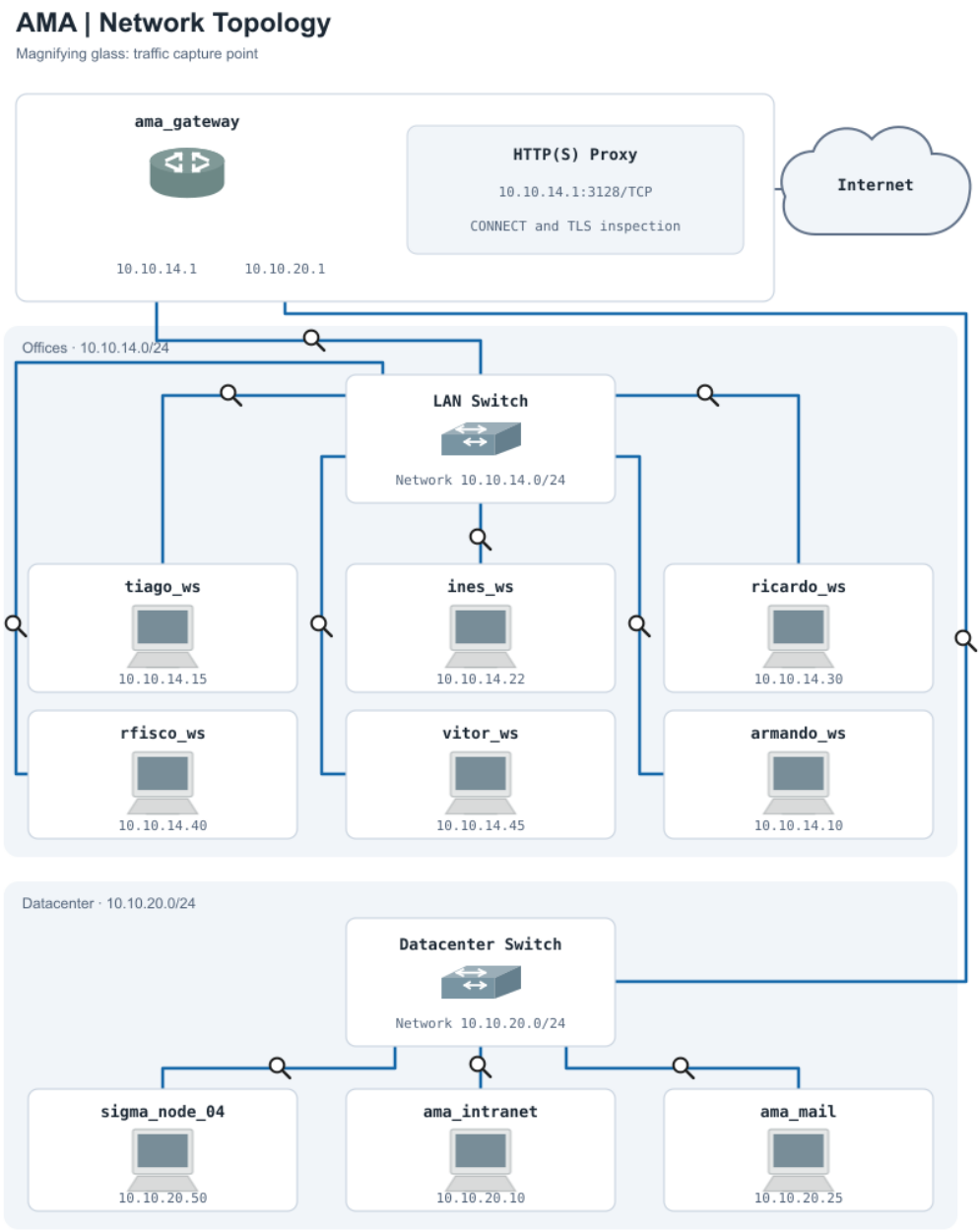


Figure 1: Overview of AMA's network.

The personnel identified in the initial case briefing include:

- **Tiago Sintaxe** - Data Science Intern and RAG Evaluator.
- **Inês Paralelo** - Information Retrieval Specialist.
- **Ricardo Vale** - Audit and Cybersecurity.
- **Rui Fisco** - Senior Data Engineer.
- **Armando Sacos** - Technical Director.
- **Vítor Algoritmo** - Cloud Operations Lead.

For auditing purposes, network traffic from AMA's infrastructure was captured during selected periods. You are provided with three network traces recorded at different times, together with the corresponding TLS key log files. Loading these key log files into Wireshark enables decryption of the captured TLS sessions for which keys are supplied, including HTTPS and other protocols carried over TLS.

As part of AMA's enterprise network architecture, selected outbound HTTPS connections are routed through an HTTP CONNECT proxy with TLS inspection to enforce security policies and monitor external access. The captures may therefore show separate connections between clients and the proxy, and between the proxy and destination servers.

File	SHA-256 digest	Description
`t1.pcapng`	`744ec38ae88e74b0f4febf9f807df10f452fb5acbc32bdbca64ac57d016a1153`	Network trace 1
`t2.pcapng`	`46b1de4fc045fa919184a8f96b0abe611c8ab6f39a146f19c15d2e39ec753ce3`	Network trace 2
`t3.pcapng`	`cfdcca3c4ccf18f855b26bce941e2b7bb94ea770cad74da0e5c6e8a492955e11`	Network trace 3
`t1-tls.keys`	`626097fd95c567c94c61247048a92da0638ab8b8fd557f0ba71e9d7698eaa8d4`	TLS key log for trace 1
`t2-tls.keys`	`1e9fe42fbb100e0ccfb195cab416c1550d2b056e207d00eb5b9f17fb2b147cf3`	TLS key log for trace 2
`t3-tls.keys`	`46cf5f261d306bdbbc81310647f0b4fa65d20e26f67a49d7bb41897941d75002`	TLS key log for trace 3

Table 1: Provided network traces and TLS key log files with their SHA-256 digests.

All relevant files are available under *Course Material* > *Lab Assignments*, through the ZIP download link next to this assignment statement.

Forensic Assignment

Investigation and evidence analysis. Conduct an impartial examination of the available evidence and document your investigation step by step, including the tools and methods used, the hypotheses tested, and all relevant findings. Support your conclusions with references to packets, timestamps, and recovered files, providing sufficient detail for your analysis to be understood and, where applicable, reproduced. Consider evidence provenance, integrity, and temporal context, distinguish observations from inferences, and state any limitations.

Questions to answer. Based on the evidence examined during the investigation, your report must explicitly answer the following questions. Support each material conclusion with identified evidence, clearly distinguishing directly observed facts from forensic inferences and hypotheses.

1. Which of the eight hidden files identified in the previous assignments can be recovered or reconstructed from the provided artifacts, and how? Where recovery is partial or impossible, explain what evidence of the file is available and what prevents its complete recovery.

2. Based on the observed network activity, what can you deduce about the identity of the person or persons responsible for transferring the documents?

3. What relevant activities involving AMA's systems can you identify in the traces? Explain the tools, techniques, and protocols involved, their apparent purpose, and their observed results. Examine how access was attempted or obtained, including the use of credentials and configuration data, and distinguish routine activity from activity requiring further investigation.

4. What do the evaluation requests and responses reveal about AMÉLIA's behaviour? Examine any relevant configuration or software changes and assess whether their effects can be established from the available evidence. Compare your findings with the hidden files recovered in the previous assignments and the explanations given in communications.

5. What path did the hidden files take from their earliest evidenced sources to João Musk's possession and their subsequent concealment and publication? Explain the connections between the relevant events, hosts, accounts, and communications across the three assignments.

6. What conclusions does the combined evidence support about the allegations that prompted Operation DeepFISH? Assess the possible responsibility of the relevant individuals, considering evidence both for and against their involvement and distinguishing the use of a host or account from

attribution to a person. Explain which conclusions are sufficiently supported, which remain unresolved, and what further investigative steps you recommend.

Note: The material provided for this assignment represents a limited view of a network environment prepared for forensic analysis. Some aspects have been simplified for teaching purposes and may differ from those found in an operational deployment. The events, characters, and allegations form a fictional scenario and should not be interpreted as factual claims about any real individuals or institutions.

Deliverables

Submit a single ZIP archive through Fenix by **21 October 2026**. The archive must contain the following four deliverables:

- **Digital forensic report.** A report documenting the investigation and evidence analysis performed. The report must explicitly answer all case questions. Each material finding or conclusion must be supported by identified evidence and by a sufficiently documented and reproducible analysis process. Use the report template available on the course website unless instructed otherwise.
- **Evidence artifacts.** The relevant evidence artifacts recovered or preserved during the investigation. The report must identify each submitted artifact by file name and SHA-256 hash, where a stable byte-exact representation is available, and state its source or provenance.
- **Auxiliary items.** Any scripts, programs, command outputs, notes, or other supporting material required to reproduce or substantiate important findings.
- **Timeline.** A timeline of the most relevant events in the case. Each event must be supported by evidence and identify the source of its timestamp, together with any assumptions or time-zone conversions applied. Use the timeline template available on the course website unless instructed otherwise.

Use of Artificial Intelligence

The use of Artificial Intelligence (AI) tools is permitted in this assignment, provided that it complies with the applicable Instituto Superior Técnico guidelines for the responsible use of AI.

Students remain fully responsible for all submitted work, including findings, technical conclusions, scripts, commands, procedures, and any other material produced with the assistance of AI tools. AI-generated or AI-assisted content must be independently verified for correctness, validity, and relevance.

Students must understand and be able to explain, justify, and, where requested, reproduce or demonstrate the investigation they submit, including the procedures followed, the tools and techniques used, the evidence obtained, and the conclusions presented. Failure to do so may be taken into account when assessing whether the student has demonstrated the competencies evaluated by this assignment.

Any use of AI tools must be declared in accordance with the applicable Técnico guidelines, identifying the tools used and the purposes for which they were used. Students should consult the guidelines and regulations published by the Conselho Pedagógico at: https://conselhopedagogico.tecnico.ulisboa.pt/regulamentos/faqs-regulamento-de-avaliacao-2026/

The Fundação para a Ciência e a Tecnologia (FCT), through FCCN, also provides the **IAedu** platform to the Portuguese higher-education and research community. The service is available to Técnico students and provides access to several AI models through a single portal. Students wishing to use AI tools in this assignment are encouraged to use this institutional service: https://si.tecnico.ulisboa.pt/servicos/servicos-externos/iaedu/

Good luck!